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Data Analytics and Data Science
Research Assignment 1

Question 1

A database is an organized system used for collecting, storing and managing data so that it can be easily accessed, updated and analyzed.

Two real world examples of where databases are used to store student personal information, course information and staff details.

1. University Systems - University Systems use databases to store student personal information, course information and staff details.

2. Banking Systems - Banks use databases to store customer information, transaction histories, account details, balances and loan records.

Question 2

Datawarehouse collects data from different sources and stores them in a central ~~collection~~ location that is easy to query and analyze.

How it differs from regular database.

Regular databases are used for daily business operation or transactions and Datawarehouse are designed to query large volumes of data from different systems.

Why would a company use data warehouse

Companies may use datawarehouse to manage large volume of data that is produced daily and combine data from multiple systems to make business decisions.

Question 3

A Datalake Stores both relational and non-relational data from different sources, data is stored in its original format until it's needed.

How it differs from datawarehouse
Datalake Stores structured and unstructured data, data is stored in its original raw format and Datawarehouse mainly stores structured data that can be used for business analytics, reporting and dashboards.

When would a business prefer a datalake over a datawarehouse.

Business would prefer datalake when massive or large amount of data needs to be stored from different sources and when the business plans to use machine learning or artificial intelligence.

Question 4

Data mart is a subset of a datawarehouse designed to serve the needs of a business unit or group of users within an organization to improve team efficiency and reduce costs.

How does it relate to data warehouse:

Datawarehouse stores data for the entire organization it contains company wide information, And Data mart stores only a portion for that data for a specific department.

who typically uses it.
Different departments of the organization uses data, mart, such as Sales team, financial team or HR.

Question 5.

Difference between Structured, Semi-Structured and Unstructured data.
Structured data → IS data that is organized in rows and columns in a format.
Example of Structured data: Spreadsheet with organized rows and columns.

Semi-Structured data → Semi structured data does not follow a fixed schema but still includes markers or metadata that defines relationships, this data sits between structured and unstructured data format.
Example of Semi-structured data is XML or JSON

Unstructured data → Data that is not organized, it does not have a predefined format and includes content such as images, videos, audio files and text documents.

Example of Unstructured data: A video recording.

Question 6.

Data Modeling - is the way of organizing data in a structured way so that it can be stored, managed and used in a system.

Why is it important when designing a system that works with data.
→ It avoids unnecessary duplication of data and also reduces error and inconsistency

Two types of data Models

Conceptual data model → Is the high level overview of the data used in organizations.

Logical data Models → It is more detailed than the conceptual model with technical information such as defined structure, organization and relationship of the data.

Question 7

Data mining is the process of using ~~statistical~~ statistical methods and machine learning to discover patterns and insights in large datasets.

How is it used in business

Business use data mining to turn raw data into valuable knowledge for decision making to help companies to understand customer behaviour.

Two real world examples of data mining in action.

• Retail → Supermarkets or stores use data mining to find which products are often bought.

• Banking → Banks use data mining to discover and detect unusual or suspicious transaction.

Question 8

Database Management System (DBMS) is used to create, store, manage and retrieve data in a database.

four popular DBMS platforms
MySQL is a relational database management system that uses structured data stored in rows and tables.
Use Case: web application such as e-commerce sites or blogs.

NOSQL → Stores data in different database formats such as documents and ~~graphs~~ graphs.
Use Cases: Application with unstructured and rapidly changing data like social media platforms.

Oracle database → Stores data as objects while maintaining relationships between them.
Use Cases: Large organizations with critical systems.

PostgreSQL → Is an object relational database management system known for advanced features like complex queries, data integrity and support for JSON.

Use Cases: Systems that need strong data consistency.

Question 9

ETL is the process of combining data from multiple sources into a large central repository called a data warehouse.

ETL Steps.

Extract → Data is collected from different systems such as databases, Excel files or cloud applications and converted into consistent formats so that it can be analyzed.

Transform →

Load → Data is now stored in a system usually a data warehouse or data mart.

Why is the process important in data analytics?
ETL is important because it combines data from various sources into a central system, improves data quality, ensures consistency in data and improves decision making.

Question 10

CSV → Is a simple text format in which data is stored in rows and columns separated by commas. I would use it in Excel and basic reporting tools because it is easy to read and use.

XML → Is a text-based file format used to store and transport structured data in a readable way. It would be used in web service or enterprise systems where data structure needs to be strictly defined.

JSON → is a semi-structured format that stores data in key-value pairs.

Question 11

Artificial intelligence is a field of computer science that focuses on creating machines and systems that can perform tasks that require human intervention.

Narrow AI are designed to perform limited set of tasks and General AI can understand, learn and apply intelligence across any task.

Examples of each

Narrow AI → Chatbots for customer service.

General AI → A human Robot.

Question 12

Machine Learning is a way to train a Software called a model to make predictions or generate content using data.

How does it differ from traditional programming?

In traditional programming humans write rules and instructions and in Machine Learning systems learn the patterns from data.

3 Main types of Machine Learning

Supervised → Is when a model is trained using data with known inputs and outputs.

Unsupervised → Uses unlabeled data and the system finds hidden patterns.

Reinforcement Learning → Models make predictions by getting rewards or penalties based on actions performed within the environment.

Question 13.

Deep learning is a subset of machine learning that is driven by many layers to learn patterns from large amount of data.

How does it relate to machine learning and artificial intelligence.

Machine learning is the subset of artificial intelligence that enables a system to learn and improve without being coded explicitly and machine learning artificial intelligence is concerned with building computers and machines that can reason and act like a human being and therefore deep learning is then a specialized subset of machine learning which is a subset of artificial intelligence.

Neural Networks and how they work.

Neural Networks are systems that are inspired by human brains, they consist of input layers and hidden layers.

How they work → You teach the system / the computer how to recognize objects by showing it several pictures of a certain object and the network then automatically learns all the features that makes up that object.

Question 14.

Natural Language Processing → Is the technology that allows or helps computers to interpret, understand and generate human language, they allow computers to read, listen and respond in a way that feel natural to a human. (3 world examples and AI's that responds with voice.)
Voice Assistance and AI's that responds with voice.

Question 15

Large Language Models (LLM) are a category of deep learning models that are trained on large amount of data, making them possible to provide responses like that of a human being.

How is it different from a standard machine learning model?

Standard machine learning ~~and~~ models are designed for smaller tasks such as predicting house prices or credit scoring while LLM are designed for performing wide range of tasks such as chatbots, translation and content creation.

Two well known LLMs and what they are used for.

ChatGPT ~~→ they are~~ used for answering questions, writing essays, coding assistance and reading documents.

Language Translation and Sentiment analysis ~~→~~ Used for highly accurate translations for different languages and analyze the underlying emotional tone of text.

Question 16

Generative AI is a type of artificial intelligence that create content such as text, images, music, videos and computer codes based on the pattern it learns from data that already exist, they can respond naturally to human conversations.

How it works at a high level

The model is trained using large amount of data in books, images, code or articles, it learns grammar, style, structure, patterns from the training data, receiving prompts and generating content.

3 tools that use generative AI and what they do

- Copilot > for Microsoft products integration, it can draft, summarize and answer questions.
- Claude > Process more data at a time and it is best for quality writing and you can create web apps.
- Wix > for building websites.

Question 17

A training dataset is the collection of data that is used to teach AI models in order for them to perform tasks.

The importance of the quality of data training data
when building AI models.

High quality data ensures the AI learns correctly and produce accuracy, fairness and reliability.

What could go wrong if the training data is biased or incorrect?

This could lead to inaccurate predictions and decisions, misleading people and cause serious legal or social issues.

Question 18.

A Computer vision enables machines to analyse, interpret and pull meaningful data from images and videos, replicating human sight and cognitive abilities.

How

Machines see and replicate images using devices like cameras, drones or scanners to record an image or video then analyse and make sense of data.

Two real world use cases of Computer vision

Security → Machines can monitor premises through cameras and identify suspicious activities or detect unusual activities and notify the owner or authorities.

Healthcare → Computer vision can help with disease identification and do diagnoses by analysing x-ray images or MRI.

Question 19

Application Programming Interface (API) are sets of rules or protocols that enable software applications to communicate with each other.

Why APIs Important in modern software and data systems.

APIs allows seamless integration and connectivity and controls what data is exposed by allowing authentication keys to ensure that authorized requests get through.

Question 20

An API key is a ~~key~~ code used to identify and authenticate an application or user. The purpose of an API key is to authenticate requests, control access and control access.

How should it be stored and protected: API keys should be protected in the same way as passwords rather than directly into source code, using secret management services.

What could happen if it is exposed publicly. If it is exposed, there could be unauthorized access to associated resources, attackers may use the key to make requests and there could be data breaches.

Question 21

REST API is an application programming interface that follows the design principles of the REST architectural style, it is a set of rules and guidelines about how you should build a web API.

REST stands for Representational State transfer

The main HTTP methods are

GET → Used for requesting information from the server.

POST → Used for sending data to the server to create a ^{resource}.

PUT → Used to send data to a server to ^{update} a resource.

DELETE → Used to delete the specified resource.

PATCH → Used to apply partial changes to a resource.

Question 22

JSON is a text base format for storing and exchanging data in a way that is readable to humans and machine-parseable. Why it is commonly used when working with APIs and data exchange is because its language independent nature makes it an ideal format for exchanging data across different programming languages and platforms.

Simple example of JSON object:

```
<1> JSON
{
  "Name": "Tendin",
  "age": 25,
  "Student-Number": 12345
}
```

Question 23

Prompt Engineering is the way of designing effective inputs to guide the AI system towards accurate and useful outputs. Prompt Engineering is important when working with AI tools like ChatGPT or Claude because it influences the quality, relevance and accuracy of outputs and the quality, relevance and accuracy of outputs.

Produce meaningful prompts	Improved version
Poorly written prompts	"What is JSON and what is it used for?"
"Tell me about ML"	"Explain what machine learning is and give me real world examples"

Question 24

MAP function executes a specific function for each item in an iterable or array.
→ It allows to apply a particular function on every element in an array and returns a new array with modified elements.

A simple example in JavaScript
`const colours = ["green", "red", "yellow"];`

Question 25

Cloud computing is the delivery of computing services such as servers, storage, databases, networking and intelligence over the internet.

3 main cloud service models.
Infrastructure as a Service (IaaS) → Offers access to computing resources such as servers, storage and networks.

example: Virtual machines service

Platform as a Service (PaaS) → Provides ready to use environment such as Microsoft Azure App Service, that allows developers to code, test and deploy applications.

example: Google App Engine.
Software as a Service (SaaS) → Provides fully developed applications over the internet, No set up is required.

example: Microsoft 365.

Question 26

A version control system is a system that is used to track and manage changes to a software code.

What is Git and why is it widely used by data analysts and developers?

Git is a distributed version control system that helps track and manage changes in a code over time. It is widely used by data analyst and developers because it allows collaboration on the same project, viewing of previous changes and restoring of changes when needed.

A commit is like a save point in a project.

A Branch is a movable pointer that refers to a specific commit.

Question 27

A Jupyter Notebook is a web application that allows data scientist to create and share documents.

The programming language it is commonly used with is Python.

It is popular because it combines code and explanation, making it simpler to visualize data, provide reporting and storytelling with data.

Question 28

Python is a programming language that is widely used in Software development, web application, machine learning and data Science.

It is considered popular programming language for Data Analytics and AI because of its simple Syntax, it is faster to write and understand with good data handling capabilities and it integrates AI and machine learning tools to support advanced AI models.

A Python Libraries commonly used

- NumPy > Offers comprehensive mathematical functions. Supports arrays and has functions for working in domain of linear algebra.
- Matplotlib > Used for data visualization to create graphs.
- Scikit-learn > Is a simple and efficient tool for predictive data analysis.
- Pandas > Is a fast powerful and flexible data analysis and manipulation tool which is easy to use when working with data sets.

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Question 29

Data Privacy is the protection of personal information and making sure that people have control over how their data is collected, stored and used.

It is important because it prevents misuse of personal data, ensures to protect people's rights and allows legal compliance.

General Data Protection Regulation (GDPR) is Europe's ~~data~~ data privacy and security law and it applies to European Union countries.

Question 30

Data Ethics are moral principles and guidelines that govern how data is collected, stored and used to protect personal information and people's privacy.

3 examples of unethical use of data or AI

- Unauthorized use of personal information → When a company collects people's email, location or browsing histories and share/sells it with third parties without a person's consent which can lead to identity theft or fraud.

- Misinformation → When AI is used to create fake news and videos that look realistic and can lead to reputational damages, fear and influence violence.

- Unintended harm → When AI Systems are not designed or tested properly where it can lead to individuals or society harms in case of treatment recommendations.

Question 31

Data Governance are frameworks of policies, roles processes and technologies that ensures that data is managed and protected and it is used responsibly.

Data Quality > Measures how well datasets meet criteria for accuracy, validity, consistency and fitness for purpose.

Data Cataloging > Provides centralized metadata repository for organization's data assets.

Data Classification > Involves organizing and categorizing data on its criticality, ~~sensitivity~~ and value.

Data Security > Provides protection of data against unauthorized access

Question 32

Bias in AI refers to discrimination embedded in AI systems resulting in unfair or harmful results.

An AI model can be biased by struggling to accurately identify people with darker skin tone, which can lead to discriminatory outcomes. Potential consequences can be denied jobs, false arrest and loss of trust in technology.

Question 33

Roles	Skills	Tools
Data Analyst	Data visualization	PowerBI, Tableau
Data Scientist	Data modelling & ml	Python, Spark, Data
Data Engineer	Database Management	PostgreSQL, AWS
AI Engineer	Algorithms & Model deployment	APIs, Cloud services

Question 34

Business intelligence uses business analytics, data mining, data visualization and data tools to help organizations make better data-driven decisions.

Two popular BI tools

Microsoft PowerBI > Connects multiple data source and helps businesses to monitor business performance quickly with interactive reports & dashboards

Tableau > Helps people to see, understand and act on data, businesses can identify trends, patterns and act as needed.

Question 35

The Artificial Intelligence and Machine Learning concept was interesting for me, I had little knowledge on the concept and realize it is much broader and kind of gives an idea of how the future is transforming technologically.

Question 36

APIs, Tools and Developer Understanding but using website like W3Schools, IBM, Snowflake and Microsoft learn to compare the meaning helped me understand more better. I also used Co-pilot to actually take the meaning to simpler terms which really helped with understanding.

Question 37

With the data that can be fed to AI and allow AI to do the work that humans would do is a bit scary but I think it would affect me if I don't learn how to use AI so that it works in my favour. for examples if chatbots are resolving user errors or answering questions that users might have, then there will be no need for me to actually resolve issues but if I can learn how to actually use AI to resolve issues then it's much better.

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